

summer 25 [B]

Q7 In a Go-Back-N TCP connection, the client and server have the following values and flow:

ISN of Client = 1100, RWND of Client = 8000;

ISN of Server = 1214, RWND of Server = 3200

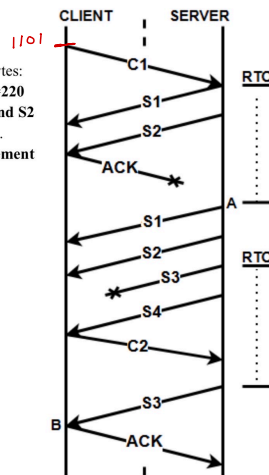
The data size of the segments is given as shown in bytes:

C1=?, C2=132, S1=1260, S2=1443, S3=122 and S4=220

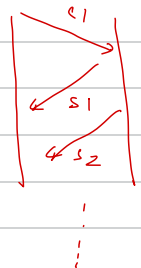
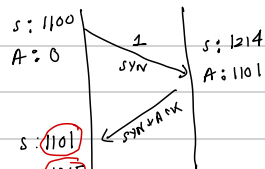
I. If the acknowledgement number of the second S2 segment is 2000, calculate the data size of C1.

II. Calculate the sequence and the acknowledgement number of the C2 segment.

III. Calculate the rwnd of the client at point B.



3
+
4
+
3



i. s_2 is sent from server immediately after s_1 .

So, Ack. number of s_2 is the same of $s_1 = 2000$

Seq. number of client = $1101 + 1(\text{TCP}) = 1101$

$\therefore$ Ack of $s_2 = \text{Seq. of client} + C_1$

$$2000 = 1101 + C_1$$

$$C_1 = 899$$

ii. Sequence number of $C_2 = 1101 + \text{Send}(C_1)$

$$= 1101 + 899$$

$$= 2000$$

Ack num of $C_2 = 1215 + \text{Receive}(S_1, S_2)$

$$= 1215 + 1260 + 1443$$

$$= 3918$$

iii. $\text{rwnd} = (8000 - S_1 - S_2 - S_3) = 5175$

Summer 25 [A]

Q7 In a **selective repeat** TCP connection, the client and server have the following values and flow:

ISN of Client = 6729, RWND of Client = 3200;

ISN of Server = 6214, RWND of Server = 8000

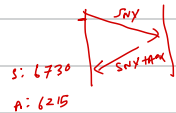
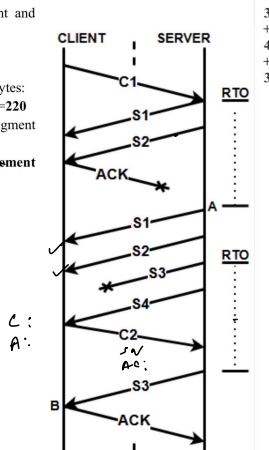
The data size of the segments is given as shown in bytes:

C1=1421, C2=1320, S1=7, S2=443, S3=122 and S4=220

I. If the **sequence** number of the **second S2** segment is 7000, calculate the data size of S1.

II. Calculate the **sequence** and the **acknowledgement** number of segment C2.

III. Calculate the rwnd of the client at **point B**.



$$i. \text{ seq. number of } S_1 = 6214 + 1 = 6215$$

$$\text{seq. number of } S_2 = \text{seq. } S_1 + \text{size of } S_1$$

$$7000 = 6215 + S_1$$

$$S_1 = 785$$

$$ii. \text{ seq number of } C_2 = 6730 + \text{size of } C_1 [\text{send}]$$

$$= 6730 + 1421$$

$$= 8151$$

$$\text{Ack number of } C_2 = 6215 + S_1 + S_2 [\text{received}]$$

$$= 7443$$

$$iii. \text{ rwnd of client at B} = 3200 - S_1 - S_2 - S_3 - S_4$$

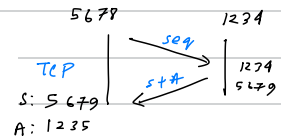
$$= 1630$$

Spring -25 [A]

Q8 After the TCP handshaking, the client with **RWND=10000 bytes** and the server with **RWND=20000 bytes** transferred the following segments using **selective-repeat** protocol. The ISN of the client is **5678** and the ISN of the server is **1234**.

From	To	Segment Name	Data Size (bytes)	Lost
Client	Server	C1	546	No
Server	Client	S1	786	No
Server	Client	S2	685	Yes
Server	Client	S3	256	No
Client	Server	Ack-1		No
Server	Client	S2	685	No
Client	Server	Ack-2		No

- [CO3] I. Calculate the sequence and the acknowledgment number of the Ack-2 segment sent by the Client.
 [CO3] II. Calculate the receiving window size of the Ack-1 segment.
 [CO3] III. Calculate the server's Sf value after sending the second S2 segment.



i. Sequence number = Seq. of client + C1[send] A:

$$= 5678 + 546$$

$$= 6225$$

Ack number = Ack. of client + S1 + S3 + S2

$$= 1235 + 786 + 256 + 685$$

$$= 2962$$



Sn = send next

Sf = send first

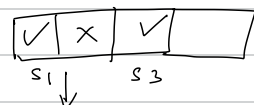
ii. Receiving window size of Ack-1 = $10000 - 786 - 256$

$$= 8958$$

iii. After sending S2 segment that are not acknowledged.

So, Sf value = 1235 + 786

$$= 2021$$



$$Sf = 1235 + S1$$

Spring 25 FBJ

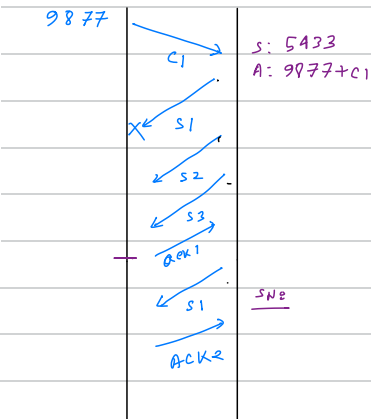
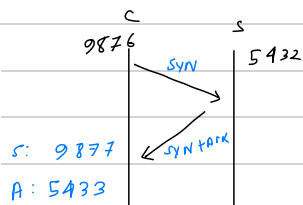
Q8 After the TCP handshaking, the client with **RWND=8000 bytes** and the server with **RWND=16000 bytes** transferred the following segments using **Go-Back-N** protocol. The ISN of the client is **9876** and the ISN of the server is **5432**.

From	To	Segment Name	Data Size (bytes)	Lost
Client	Server	C1	645	No
Server	Client	S1	687	Yes
Server	Client	S2	586	No
Server	Client	S3	652	No
Client	Server	Ack-1		No
Server	Client	S1	687	No
Client	Server	Ack-2		No

- [CO3] I. Calculate the sequence and the acknowledgment number of the Ack-1 segment sent by the Client.
 [CO3] II. Calculate the receiving window size of the Ack-2 segment.
 [CO3] III. Calculate the server's Sn value after sending the second S1 segment.

THE END

2



a. sequence number = $9877 + C_1$ (sent)
 $= 9877 + 645$
 $= 10522$

Ack. number = $5433 + \text{receive}$ [nothing receive as Go-back protocol client drop until it receive S1]
 $= 5433$

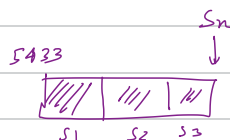
b. After ack-2, client received S1 only.

receiving window size = $8000 - 687$ [S1]
 $= 7313$

Sn = send next

c. After send second S1 segment, it will point to the next new byte to send

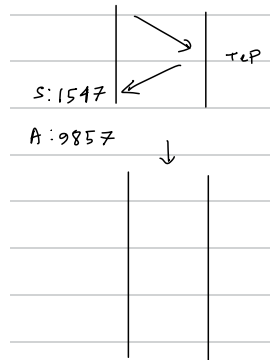
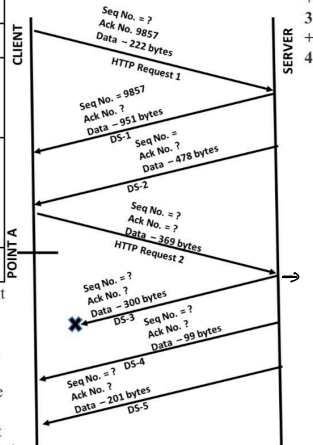
$= 5433 + S_1 + S_2 + S_3$
 $= 7358$



Fall-22 [B]

c) In a **selective-repeat** TCP connection, the client & server have the following values and flow:

	Client	Server
ISN	1546	9856
Segment Sizes DS = Data Segment	HTTP Request 1: (Also the third segment of the 3 way handshake): 222 bytes	DS 1: 951 bytes DS 2: 478 bytes
	HTTP Request 2: 369 bytes	DS 3: 300 bytes DS 4: 99 bytes DS 5: 201 bytes
RWND	1684	6599



- Calculate the sequence and acknowledgment number of the server's Data Segment 3 sent to the client.
- The client sends all the received segments to the upper layers at point A. Determine the rwnd of the client after the client receives the 5th Data Segment.
- What are the sequence and acknowledgment number of the acknowledgment segment that the client will send (not shown) after it receives the 5th Data segment?

S:

$$\begin{aligned}
 \text{i. Sequence number} &= 9857 + \text{DS1} + \text{DS2} \quad [\text{sent}] \\
 &= 9857 + 951 + 478 \\
 &= 11286
 \end{aligned}$$

$$\begin{aligned}
 \text{Ack. number} &= 1547 + \text{HTTP 1} + \text{HTTP 2} \quad [\text{receive}] \\
 &= 1547 + 222 + 369 \\
 &= 2138
 \end{aligned}$$

ii. After point A, all sent to Application layer.

$$\begin{aligned}
 \text{So, new rwnd after 5th} &= 1684 - \text{DS4} - \text{DS5} \\
 &= 1684 - 99 - 201 \\
 &= 1384
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. Sequence number} &= 1547 + \boxed{\text{req 1} + \text{req 2}} \quad [\text{sent}] \\
 \text{from client} &= 1547 + 222 + 369 \\
 &= 2138
 \end{aligned}$$

$$\begin{aligned}
 \text{Ack number} &= 9857 + \text{DS 1,2} = 9857 + \boxed{951 + 478} = 11286
 \end{aligned}$$

Full - 22 [A]

Q 3. a) The UDP header has a field called checksum. **Explain** what this field does.

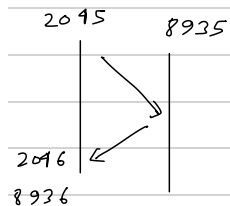
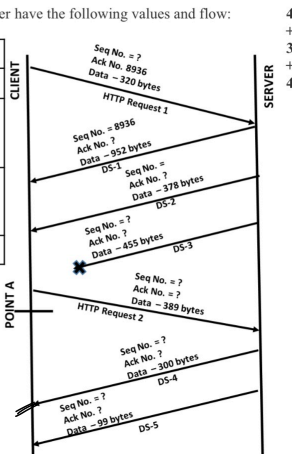
C03

b) PCA sends data of 1800 bytes starting from sequence number 3001 to PCB. PCA wishes to indicate that 700 bytes of data have to be processed urgently. **Discuss** how PCA will inform this to PCB.

c) In a **go-back-n** TCP connection, the client & server have the following values and flow:

	Client	Server
ISN	2045	8935
Segment Sizes DS - Data Segment	HTTP Request 1: <i>(Also the third segment of the 3 way handshake):</i> 320 bytes HTTP Request 2: 389 bytes	DS 1: 952 bytes DS 2: 378 bytes DS 3: 455 bytes DS 4: 300 bytes DS 5: 99 bytes
RWND	3020	6450

- Calculate the sequence and acknowledgment number of the server's HTTP Request 2 sent to the server.
- The client sends all the received segments to the upper layers at point A. **Determine** the rwnd of the client after the client receives DS-5?
- What are the sequence number and acknowledgment number of the client after receiving DS-4?



$$\begin{aligned}
 \text{1. seq. number} &= 2046 + \text{req 1} \\
 &= 2046 + 320 \\
 &= 2366
 \end{aligned}$$

$$\begin{aligned}
 \text{ACK number} &= 8936 + \text{DS 1, 2} \\
 &= 8936 + 952 + 378 \\
 &= 10266
 \end{aligned}$$

ii. After point A, all sent to Application layer.

As used go-back-n protocol after DS-3 noting receive.

Now, rwnd = 3020 [go-back-n]

$$\text{iii. seq. number} = 2046 + \text{req 1} + \text{req 2} = 2755$$

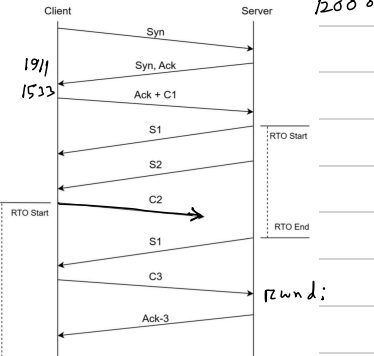
$$\begin{aligned}
 \text{ACK. number} &= 8936 + \text{DS 1} + \text{DS 2} \quad [\text{DS 4 not receive}] \\
 &= 10266
 \end{aligned}$$

Fall-24 [A]

Q8 While transferring data, the ISN of the client is **1910**, and the ISN of the server is **1532**. The data size of the segments are given as shown in bytes:

C1=421, C2=320, C3=111, S1=260, and S2=220.

Initially, the rwnd of the client is **6000** bytes, the rwnd of the server is **12000** bytes, and they are using **selective repeat** protocol to send data.



- [CO2] I. Explain why the server is resending segment S1 and what the client will do with segment S1.
- [CO3] I. Calculate the sequence and acknowledgment numbers of the C2 segment.
- [CO3] II. Calculate the rwnd of the server after receiving the C3 segment from the client.

$$\begin{aligned}
 \text{ii. seq number} &= 1911 + C_1 \\
 &= 1911 + 421 \\
 &= 2332
 \end{aligned}$$

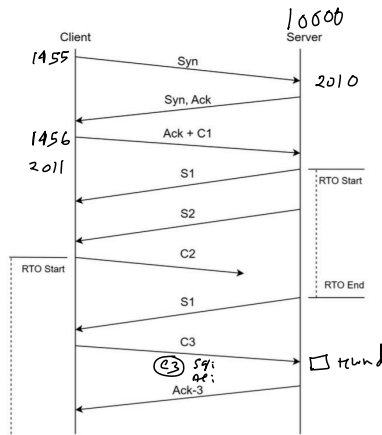
$$\begin{aligned}
 \text{Ack number} &= 1533 + S_1 + S_2 \\
 &= 2013
 \end{aligned}$$

$$\text{iii. rwnd} = 12000 - C_1 - C_2 = 11468$$

Fall-24 [B]

Q8 While transferring data, the ISN of the client is 1455, and the ISN of the server is 2010. Initially, the rwnd of the client is 8000 bytes, the rwnd of the server is 10000 bytes, and they are using go-back-n protocol to send data. The data size of the segments are given as shown in bytes:
C1= 320, C2=111, C3=260, S1=220, and S2=421.

- [CO2] I. Explain why the server is resending segment S1 and what the client will do with segment S1.
- [CO3] II. Calculate the sequence and acknowledgment numbers of the C3 segment.
- [CO3] III. Calculate the rwnd of the server after receiving the C3 segment from the client.



$$\begin{aligned} \text{II. seq number} &= 1456 + c_1 + c_2 \\ &= 1456 + 320 + 111 \\ &= 1887 \end{aligned}$$

$$\begin{aligned} \text{Ack number} &= 2011 + s_1 + s_2 \\ &= 2011 + 220 + 421 \\ &= 2652 \end{aligned}$$

$$\text{iii. rwnd} = 10000 - c_1 = 10000 - 320 = 9680$$